



STEP Interface Functional Specification

STEP interface with Integrated Assessment
System (IAS) to Push Marks

Version 0.2



Document Version

Version	Effective Date	Summary Changes	Created/ Updated by
0.1	25/03/2024	Initial version of Integrated Assessment System IAS	AvePoint
0.2	5/04/2024	Revised comments based on ITE's feedback	AvePoint
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Document Acceptance Form

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1. Introduction

1.1 Purpose of this Document

This Interface Functional Specification (IFS) is to provide the agreed baseline of communication requirement and detailed technical specification between the SaaS Skills Training & Enhancement Portal (STEP) and Institute of Technical Education (ITE) Integrated Assessment System (IAS) for send over students' exam result and module information via API.

1.2 Intended Audience

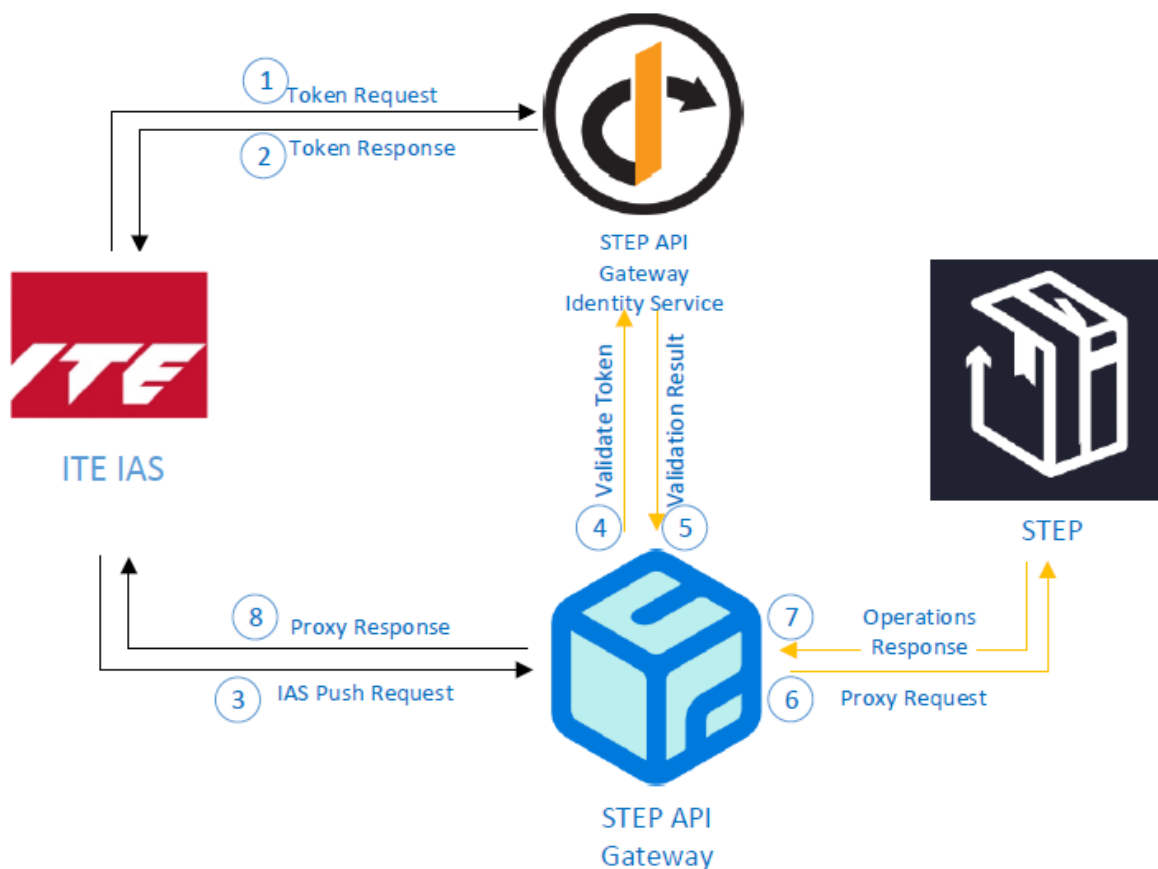
This technical document is used for system communication. Therefore, the intended audiences are the technical stakeholders, business users and system owners of both systems.

2. Technical Overview

2.1 System Flow

There are 4 roles involved in the interface diagram.

- **ITE IAS:** Institute of Technical Education Integrated Assessment System.
- **STEP API Gateway:** STEP interface Integration middleware that validates HTTP request authentication, authorization, and payload inspection.
- **STEP API Gateway Identity Service:** STEP interface Identity Issuer that issues access token for API authentication and authorization.
- **STEP:** Skills Training & Enhancement Portal.



SN	Description
1	ITE IAS sends POST request with <i>client_id</i> and <i>secret</i> to STEP API Gateway Identity Service to get API access token.
2	STEP API Gateway Identity Service sends the access token back to ITE IAS within response message. This access token can be reused within 1 hour, client application can use an access token for multiple API request before the access token expires, after that client application needs to request a new access token again.
3	ITE IAS sends HTTP POST request with IAS Result pushed results and reference date to STEP API Gateway.
4	STEP API Gateway sends request to STEP API Gateway Identity Service to validate the access token of the operation response.
5	STEP API Gateway Identity Service sends back the validation result.
6	STEP API Gateway verifies the request and sends a proxy request to STEP.
7	STEP insert IAS Result into database and send operation result.
8	STEP API Gateway sends the operation response to IAS in the attached file in. ‘Successful’ field of every record indicates if record score is uploaded successfully or not in the response body.

2.2 Interface Design Summary

Interface Name	SendIASResults
Hosting System	Skills Training & Enhancement Portal
Data Source System	Integrated Assessment System
Data Destination System	Skills Training & Enhancement Portal
Mode of Interface	HTTP API
Method	POST
Content-Type	application/json
Authorization	Bearer <access_token>
Frequency	Realtime

Encryption Method	HTTPS
Endpoint URL	<TBC>

2.3 Interface Authentication

STEP use OAuth (Oauth2.0) to perform authentication. ITE Integrated Assessment System needs to send HTTP POST request to STEP Identity Service endpoint to get access token and put the access token into HTTP request 'Authorization' to consume target API.

STEP Identity Service Endpoints:

Env	URL
UAT	https://dbp-int-uat2.stms.edutech-online.com/interface/api/ias/SendIASResults
Production	https://api.stms.polite.edu.sg/interface/api/ias/SendIASResults

To access the API, the client application needs to authenticate using the client credential grant from API gateway for an access token with the following parameters:

- **grant_type**: REQUIRED. The value must be set to "client_credentials"
- **client_id**: REQUIRED. Client ID.
- **client_secret**: REQUIRED. Client Secret.

Sample:

```
curl -location -request POST '{URL}' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--data-urlencode 'grant_type=client_credentials' \
--data-urlencode 'client_id={client_id}' \
--data-urlencode 'client_secret={client_secret}'
```

Response Body:

```
{
  "access_token": "{access token}",
  "expires_in": 3600,
```



```
"token_type": "Bearer",  
  
"scope": "{api scope}"  
  
}
```

2.4 Business Rule

2.4.1 Modules identification

Course Type	Business Rule
Semester-based course	For semester based course, it shall be identified by data fields, which include Student ID, Exam Series, Module Code and Subcomponent

2.4.2 Processing results on STEP

1. The IAS results sent over shall be recorded as "Confirmed" status in STEP, the results uploaded to STEP and amending of results would not be allowed.
For any changes to the results, CET user would have to reject subcomponent score/grade under Module Overview to change the status from 'Confirmed' to 'Unsubmitted'. Only with scores in the 'Unsubmitted' status, the original scores sent from IAS can be replaced by the subsequent uploads by user.

Course / Assessment management / Assessment processing / Module details

Nanyang Polytechnic

BM0001 - e-Business Management

Minimum attendance requirement: 60%

Grade mapping rule: Grade mapping rule 1

Sub-component weightage

CW1	CW2	Final
20%	20%	60%

Unsubmitted Pending approval **Confirmed** Reviewed All

Assessment status: All

Search by student name...

Export Grade moderation **Reject**

☐	Student name	Student ID	Course intake No.	PDC/MC code	Class	Mod
☐	VC Valentin Connolly	A001	NYP-0427-2 20046	Core 1	Class 1	97
☐	DL Declan Lehner	A002	NYP-0427-2 20046	Core 1	Class 1	97
☐	EG Eulah Glover	A003	NYP-0427-2 20046	Core 1	Class 1	93

2. For any changes to the grades/score for assessment, the user would have to complete the approval process flow in STEP and follow the cut-off date settings on STEP.

Course / Assessment management / Exam series cut-off date settings

Nanyang Polytechnic

ES3

Exam series cut-off date settings

☒ Reminder to be sent 1 day after rejection

☒ Reminder to be sent 1 day before the cut-off date

Grade confirmation cut-off date

Score entry cut-off date: 31/07/2022 18:00

Module grade confirmation cut-off date: 01/08/2022 18:00

Grade review & publish cut-off date

For the course chair to review the grade before confirming and publishing to students.

Course chair confirm score/grade cut-off date: 02/08/2022 18:00

BOE Endorsement date: 03/08/2022 18:00

Publish score/grade date: 03/08/2022 18:00

2.4.3 Email notification for each upload

An email notification will send to <ITE Admin email address> to report the records process details after STEP successfully handled OMR http request (status code of 200/ 400/ 500). Email template as follows:

Subject: YYYYMMDDhhmm IAS – STEP upload details

Attachment: YYYYMMDDhhmm IAS Upload to STEP.csv

Body:

Dear Officer,

Results have been uploaded to STEP from ITE Integrated Assessment System.

Statistics- {success count} successful, {error count} error of total {total count} records.

Possible error messages for record level are listed as follows:

SN	Error message
----	---------------

1	Invalid format for '{field name}'.
2	No '{field name}' value found.
3	Results cannot be uploaded because current status of results is "confirmed". <i>(Existence of confirmed results which cannot be overwritten unless status of results is changed from 'confirmed' to 'unsubmitted')</i>
4	Upload date later than score-entry cut-off date.

Sample attachment csv file:



20240325271627
IAS upload to STEP.csv

Should there be a status code of 400 or 500, an email template would be as follows:

Subject: YYYYMMDDhhmm IAS – STEP upload details

Attachment: YYYYMMDDhhmm IAS Upload to STEP.csv

Body:

Dear Officer,

Identified 400 – Bad Request Error during the upload process to STEP.

Please raise a support ticket to STEP support team for further investigation.



Subject: YYYYMMDDhhmm IAS – STEP upload details

Attachment: YYYYMMDDhhmm IAS Upload to STEP.csv

Body:

Dear Officer,

Identified 500 – Internal Server Error during the upload process to STEP.

Please raise a support ticket to STEP support team for further investigation.

3. Interface Fields

3.1 Interface launch parameter

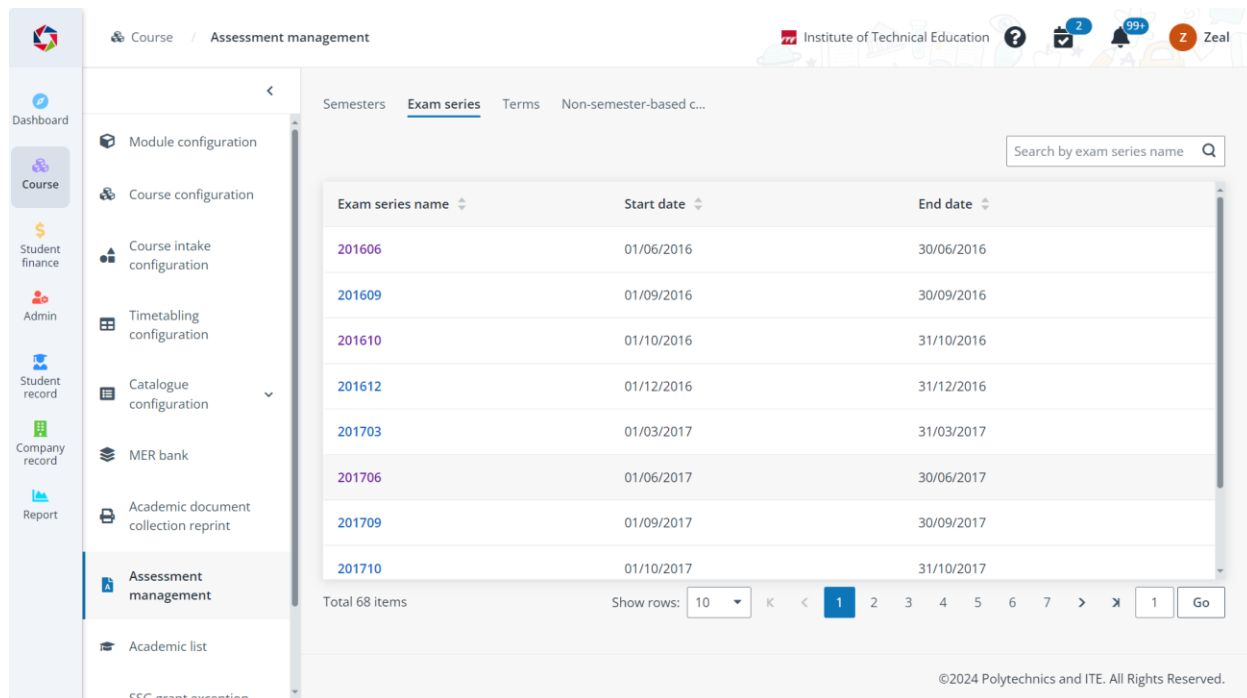
STEP Field Name	IAS Field Name	Description	Data Type	Length	Remarks
StudentId*	StudentId	UAM Common ID	String	20	[For info – these will flow from STEP to iStudent (via student records interface) to IAS, and then flow back from IAS to STEP]
ExamSeries	ExamSeries	Exam Series	String	20	Sample: YYYYMM
Subcomponent*	Assignment Item	Subcomponent type	String	20	Sample: TEXM
ModuleCode*	ModuleCode	Module code	String	255	Sample: BM0001FPR <i>*Note that this module code shall be unique for identification purposes for semester/non-semester courses.</i>
Score*	Mark	Assessment score	Number	-	Sample: 30
Assessment status	NonNumericGradeIndicator	To indicate if student was absent	String	10	Sample: AB, AWOL, ZR (if AB or AWOL shown = ABS in STEP) (If ZR, ignore it.) Can be blank
Modified	SFTPDateTime	Date of creation or modified date	Datetime	50	Sample: yyyy-MM-dd hh:mm:ss
Remarks	Entity name	To indicate source of marks	String	255	Sample: INS_Push_Marks

*represents compulsory fields to be sent.

3.2 Interface Response Data Fields

STEP Field Name	Description	Data Type	Length	Remarks
StudentId	UAM Common ID	String	20	
ExamSeries	Exam Series	String	20	Format: YYYYMM
Subcomponent	Subcomponent type	String	20	Sample: TEXTM
ModuleCode	Module code	String	255	
IntakeNo	Course intake number	String	100	
ClassName	Class name	String	100	
Successful	Upload status of current record item.	Boolean	-	True – Success False – Failure
ErrorMessage	Error message	String	255	Error message to explain why current record not uploaded successfully. Empty if upload successfully.
Timestamp	Timestamp marked by STEP	Datetime	-	

Please note that exam series would be indicated as YYYYMM.



Course / Assessment management

Institute of Technical Education

Semesters **Exam series** Terms Non-semester-based c...

Search by exam series name

Exam series name	Start date	End date
201606	01/06/2016	30/06/2016
201609	01/09/2016	30/09/2016
201610	01/10/2016	31/10/2016
201612	01/12/2016	31/12/2016
201703	01/03/2017	31/03/2017
201706	01/06/2017	30/06/2017
201709	01/09/2017	30/09/2017
201710	01/10/2017	31/10/2017

Total 68 items Show rows: 10 1 2 3 4 5 6 7 1 Go

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Course

Assessment management

Assessment processing

Module details

Edit

Calendar

Notifications

User Profile

←

CW1

Assessment status: All

Search by student name...

Student name	Student ID	Class	Score	Given score	Mapped grade	Given grade	Pass critical skill	Assessment status	Remarks
YC Valentin Connelly	A001	Class 1					☐	XM	
DL Declan Lehner	A002	Class 1	90.9	90.9	A	A	☒	DC	
EG Eulah Glover	A003	Class 2	69.1	69.1	D	D	☒		
DF Devin Fisher	A004	Class 2					☐		

Total items: 4

Show rows: 10

<

>

1

1

Go

Close

Save

3.3 Sample Responses

Successful Operation Sample (Status Code 200)

```
{
  "data": [
    {
      "StudentId": "S1234567",
      "ExamSeries": "202205",
      "Subcomponent": "CLT",
      "ModuleCode": "BM0001",
      "IntakeNo": "BM0001-1",
      "ClassName": "Class A",
      "Successful": "True",
      "ErrorMessage": "",
      "Timestamp": "2022-05-24 16:00:00",
    },
    {
      "StudentId": "S7654321",
      "ExamSeries": "@202205",
      "Subcomponent": "CLT",
      "ModuleCode": "BM0001",
      "IntakeNo": "BM0001-1",
      "ClassName": "Class A",
      "Successful": "False",
      "ErrorMessage": "Invalid field format for 'ExamSeries'.",
      "Timestamp": "2022-05-24 16:00:00"
    }
  ]
}
```



```
}
```

Bad Request Sample (Status code 400)

```
{
```

```
  "Request body is invalid"
```

```
}
```

Internal Server Error (Status code 500)

```
{
```

```
  "Internal server error."
```

```
}
```

4. Status Codes

The table below provides the detailed error listing and handling procedure. These status codes would be response to client in the Http response header.

Status Code	Message	Description	Handling Procedure
200	NA	Successful Request <i>(Please note that possible error messages would be included in csv file for each upload. Refer to Section 2.4.3 Email notification for each upload for more details)</i>	NA
400	Bad request	Request body is invalid <i>(This is a generic error code which indicates request body is not recognized, not to indicate a record level exception.)</i>	Verify the interface launch parameter in your HTTP request; Raise a ticket to STEP support team for further investigation.
500	Internal server error	Maintenance, system down	STEP support team will inform ITE system admin if schedule maintenance. If error occur during working hour, Raise a ticket to STEP support team for further investigation.